

Instructor Teaching Notes — AI in Clinical Practice

Diplomado en Inteligencia Artificial en Ciencias de la Salud

A lesson-plan script for every week: what to say, how to run each activity, and how to facilitate discussion.

How to use this document: Each week gives you a timed run-of-show for the 3-hour virtual session plus facilitation instructions for the 3-hour Virtual Clinical Sandbox (VCS) block. Script lines are suggestions, not a transcript to read verbatim — adapt to your own voice. Timing assumes a 3-hour (180-minute) virtual block; adjust proportionally if your institution runs shorter sessions.

■ MONTH 1 — ADMIN TIME SAVINGS & THE AI FOUNDATION

Week 1: Introduction to AI in Medicine

Session objective: Students can define HITL and explain why it anchors the whole course.

Opening (10 min) — say this: > “Over the next 25 weeks, you’re going to work with ambient scribes, diagnostic AI, and care-planning tools. Every single one of them shares one safety principle, and if you only remember one thing from this entire course, it should be this: Human-in-the-Loop. Today we define it, and every week after this, we’ll come back to it.”

Teaching script — HITL and ML literacy (40 min):

- Open with the adoption-speed hook: tell students that ambient AI scribes have spread through medicine faster than the stethoscope, the X-ray, or even the EHR itself — large deployments now cover millions of patient encounters. Ask: *“If a technology spreads that fast, who’s making sure it’s being used safely?”* Let the tension sit before answering: it’s the clinician, in real time, at the point of care — that’s HITL.
- Define supervised vs. generative AI conceptually (whiteboard-friendly): supervised = learns from labeled examples to predict a category; generative = produces new text/content that sounds fluent but isn’t guaranteed to be true.
- Define “hallucination” using a concrete, low-stakes example first (e.g., an AI inventing a citation), then bridge to the clinical stakes (an AI inventing a symptom, a lab value, or a negative finding).

Facilitating the VCS activity (Account setup + baseline timing, 3 hrs): 1.

Walk students through account creation for Synthea, the sandbox EHR, and the ambient-scribe trial license — do this live, screen-shared, so no one gets stuck

asynchronously. 2. Explain the “baseline exercise”: have each student manually document a short mock visit *without* AI assistance and time themselves. Emphasize this number matters later — Week 4 compares against it. 3. Circulate during the timed task; do not answer clinical-content questions during the timing window (it will bias the baseline).

Discussion facilitation (Q1.1):

- Present the vignette. Resist supplying the answer immediately — ask for a show of hands on A-E first, then have a student who chose “B” defend it before you confirm.
- Steer the conversation toward: *“The scribe error was inevitable. What wasn’t inevitable was the physician not reading the note. Where’s the fix — in the software, or in the workflow?”*

Wrap-up (5 min): Preview that next week is about *how* to talk to these tools, not just how to supervise them.

Week 2: Prompt Engineering — The “Four Rs”

Session objective: Students can apply Request, Reason, References, Revise to a clinical prompt.

Opening (5 min) — say this: > “Last week we talked about catching AI mistakes after they happen. This week is about reducing how many mistakes happen in the first place — starting with how you talk to the tool.”

Teaching script — Four Rs (35 min):

- Write the four words on the board and build the framework live by drafting a bad prompt first (“Summarize this visit”) and workshopping it as a class into a Four-Rs-compliant version.
- Emphasize “Revise” as the step students most often skip: a prompt that doesn’t invite correction is a prompt that hides its own errors.

Facilitating the VCS activity — live-prompting workshop (3 hrs): 1. Pair students up; one plays patient using a Synthea case card, one plays clinician. 2. After the roleplay, have the “clinician” write one prompt, run the scribe, then revise the prompt once based on what the output got wrong. Require them to submit both versions side-by-side. 3. Spot-check 3-4 pairs’ submissions in the last 20 minutes as a group review — this is more effective than reviewing all of them privately, since peer examples teach faster than private feedback.

Discussion facilitation (Q1.2):

- Have students vote, then ask specifically: “What’s missing from option D that’s present in B?” — force them to name the *absence* of the Revise step, not just recognize the presence of good elements.
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Week 3: Ambient Scribing in Practice

Session objective: Students can categorize scribe errors and explain the German literacy-gap rationale for structured training.

Opening (5 min) — say this: > “You now have a framework for talking to the tool. Today we stress-test it across a full clinic day — because errors don’t show up evenly. They cluster around certain kinds of visits.”

Teaching script — error taxonomy + German context (35 min):

- Present the three error types (mistranscription, omission, hallucinated detail) with one example each, ideally pulled from your own prior sandbox sessions if you have them.
- Bridge to the German-standard content by asking: *“If a physician has never been trained on what a hallucination even is, can they be expected to catch one?”* — this motivates why structured onboarding (not self-study) is the regulatory answer.

Facilitating the VCS activity — full mock-clinic day (3 hrs): 1. Assign each student 3–4 synthetic-patient roleplay encounters of increasing complexity (start simple, end with a multi-problem visit). 2. Require a “catch log” — a running list of every correction made to the AI draft, with the error type tagged (mistranscription/omission/hallucination). 3. Reserve the last 15 minutes for students to total their own catch logs — this creates their own personal baseline for Week 4’s discussion.

Discussion facilitation (Q1.3): Ask a German-context student (if any) or have someone research live: “Why would a literacy gap be treated as a systemic problem rather than an individual one?”

Week 4: Month 1 Synthesis & Checkoff

Session objective: Students can articulate why efficiency gains don’t imply trustworthiness.

Opening (5 min) — say this: > “Three weeks ago, you timed yourself documenting a visit with no AI help. Today, pull that number back up — we’re going to compare it to what you’ve been experiencing since.”

Teaching script — efficiency vs. trust (30 min):

- Have students share their Week 1 baseline vs. their Week 3 catch-log error count side-by-side. The pedagogical point: both numbers can look good at the same time, and that’s exactly the trap — time saved does not certify accuracy.
- Cite the finding that review time is “non-zero” even as burnout drops — frame this as the single sentence you want them to remember from Month 1.

Facilitating the checkoff (graded, 3 hrs): 1. Each student independently runs one full ambient-scribed encounter, corrects the note, and signs it. 2. Grade against a simple rubric: did they catch at least one seeded error (plant one deliberately if the platform allows), and does their corrected note reflect it accurately? 3. Students who miss the seeded error should repeat the checkoff before advancing — this is the pass/fail gate.

Discussion facilitation (Q1.4): Push back gently on any student who defends the resident's "I trust it completely now" framing — ask them to separate the *feeling* of trust from the *evidence* for it.

📅 MONTH 2 — AI-DRIVEN DIAGNOSTICS & THE SAFETY OVERRIDE

Week 5: Critical Appraisal of AI Diagnostics

Session objective: Students can explain the dissociation between benchmark performance and bedside reasoning.

Opening (5 min) — say this: > "Month 1 was about documentation. Starting today, the stakes go up — we're moving into diagnosis, where an AI's confident-sounding answer can be flatly wrong."

Teaching script — the dissociation (35 min):

- Present the meta-analysis finding: across 12,657 participants in randomized trials, AI-CDSS produced only a *modest* improvement in diagnostic accuracy over standard care — far more modest than marketing benchmarks would suggest.
- Contrast this explicitly with board-exam-style performance, which tends to be much higher. Ask: "*Why would a tool that's excellent at multiple-choice questions be only modestly better at real diagnosis?*" Let students reason toward the answer: real cases are messier, ambiguous, and multi-step in a way exam questions aren't.

Facilitating the VCS activity — observation exercise (3 hrs): 1. Run a diagnostic-AI demo against de-identified sample images (MIMIC-CXR) without letting students act on the output yet — this week is observation only. 2. Have students log the AI's stated confidence score next to the actual finding for each case, building their own small dataset of confidence-vs-correctness. 3. End with a quick tally: how often did high confidence coincide with a wrong answer? Save this for the Week 7 override exercise.

Discussion facilitation (Q2.4): Ask students to generate their own real-world analogy for "good at the test, weaker in practice" before revealing the AI-specific framing — this builds transfer of the concept.

Week 6: German Quality Pillars

Session objective: Students can distinguish Functionality, Robustness, and Comprehensibility failures.

Opening (5 min) — say this: > “Not every AI failure looks the same. Today we build a vocabulary for naming *which* failure you’re looking at — because the fix is different for each one.”

Teaching script — three pillars (35 min):

- Give one-sentence definitions and one memorable example each:
 - Functionality: “Does it work at all?” (e.g., the tool crashes or gives no output)
 - Robustness: “Does it keep working under stress?” (e.g., it works on clean data but breaks on messy real-world input)
 - Comprehensibility: “Can a human see why?” (e.g., it gives a correct-looking answer with zero traceable reasoning)
- Mention that explainable AI (XAI) is actively cited in the research literature as a promising way to reduce automation bias — this is why Comprehensibility gets its own pillar rather than being folded into Functionality.

Facilitating the VCS activity — case-sorting exercise (3 hrs): 1. Distribute a set of de-identified CDSS malfunction write-ups (build 8-10 short cases in advance, or draw from published case reports). 2. Have small groups sort each case by which pillar was violated, then defend their sorting to another group. 3. Debrief any disagreements as a full class — disagreement here is pedagogically useful, not a sign of failure.

Discussion facilitation (Q2.2): After revealing the answer, ask: “*Why isn’t this a Functionality problem? The tool DID work — it gave an answer.*” — this is the exact distinction students most often blur.

Week 7: The Sandbox Override Exercise

Session objective: Students can detect a seeded AI error and execute a documented manual override.

Opening (5 min) — say this: > “Today you’re going to feel automation bias happen to you in real time. That’s the point — this is a controlled environment specifically so you can feel the pull toward trusting the machine, and practice resisting it.”

Teaching script — automation bias + evidence (30 min):

- Share the finding that in one 210-participant study, false agreement with wrong AI recommendations was significantly reduced by diagnostic training and experience — but *increased* when participants perceived the system as more beneficial. Ask: “*Why would liking the tool more make you worse at catching its mistakes?*”
- Mention that current research is actively testing “behavioral nudges” as a countermeasure — this is a live, unsolved research question, not settled science, which is a good thing to be honest with students about.

Facilitating the VCS seeded exercise (3 hrs) — this is the core activity of Month 2: 1. Before class, seed 2-3 synthetic cases with a deliberately biased or

anomalous AI output (e.g., a risk score that's clinically implausible given the case details). 2. Do not tell students which cases are seeded. 3. Grade on three things: (a) did they notice the anomaly, (b) can they articulate *why* it's clinically implausible, (c) did they document a manual override rather than silently correcting it. 4. In the debrief, have students who caught the seed explain what specifically tipped them off — this becomes a shared “tell” vocabulary for the rest of the cohort.

Discussion facilitation (Q2.1, Q2.3): Push students past “I just knew it was wrong” toward naming the specific clinical detail that contradicted the AI. Vague pattern-matching isn't a teachable skill; naming the detail is.

Week 8: Month 2 Synthesis & Checkoff

Session objective: Students independently detect and override one new seeded error under observation.

Opening (5 min) — say this: > “Last week you practiced with peers watching. Today you do it alone, under observation — because that's what real practice will actually feel like.”

Teaching script — debrief near-misses (25 min):

- Anonymously share 2–3 near-miss patterns from Week 7 (with permission, or paraphrased) — normalizing near-misses as data, not shame, keeps students honest in self-reporting.

Facilitating the checkoff (graded, 3 hrs): 1. One new seeded case per student, observed live (in person or via screen-share). 2. Pass criteria: detect the anomaly, articulate the clinical reasoning, document the override correctly. 3. Students who fail should repeat the Week 7 exercise with a new case before re-attempting the checkoff.

🎯 MONTH 3 — CLINICAL CARE PATHWAYS & DATA SOVEREIGNTY

Week 9: AI-Enhanced Care Planning

Session objective: Students can verify an AI-drafted nursing care plan before finalizing it.

Opening (5 min) — say this: > “Everything we've done so far has centered on physicians. Today we prove the HITL principle isn't physician-specific — it applies just as much to nursing workflows.”

Teaching script — NLP + standardized care models (30 min):

- Walk through how a care-planning tool combines NLP-extracted problems with a model like NANDA-I, and where a nurse's clinical judgment must still intervene (matching extracted problems to the patient actually in front of

them, not just the chart).

Facilitating the VCS activity — care-plan drafting (3 hrs): 1. Provide each student a Synthea synthetic patient record; have them generate an AI-drafted care plan. 2. Pair students to roleplay a bedside reassessment; the “nurse” must revise the plan against what the “patient” actually reports, which will sometimes contradict the chart. 3. Collect both the original AI draft and the revised plan for comparison.

Discussion facilitation (Q3.2): Ask directly: “What made you change the plan? Was it something the AI could have known from the chart, or only something you learned at the bedside?” — this distinguishes correctable AI error from irreducible information gaps.

Week 10: GDPR & Data Sovereignty

Session objective: Students can explain why cross-border cloud AI use is restricted and why on-premises deployment is preferred.

Opening (5 min) — say this: > “Today has nothing to do with clinical judgment and everything to do with geography — literally, where does the data physically sit, and why does that matter legally?”

Teaching script — GDPR + sovereignty (35 min):

- Present the concrete case: using an external US-hosted API to process identifiable EU health data creates cross-border transfer risk under GDPR, which is why processing generally has to stay within EU jurisdiction.
- Give the German-market example: vendors specifically market GDPR-native infrastructure hosted in German jurisdiction, with on-premises, private VPC, and air-gapped deployment options where data never leaves the institution’s control. Even large US vendors have responded to this pressure with sovereign, EU-hosted offerings.
- Frame the “why”: this isn’t a preference, it’s a response to real legal exposure.

Facilitating the VCS activity — data-flow mapping (3 hrs): 1. Have students diagram the data flow of every VCS tool used this term: where does input go, where is it processed, where is output stored. 2. Mark every point where data would leave institutional control if this were a real hospital deployment. 3. Small-group presentations of their maps to the full class — this repetition across groups builds the pattern-recognition skill for spotting sovereignty risk in real vendor pitches later.

Discussion facilitation (Q3.1, Q3.4): Ask students to argue *both sides* briefly — what would a vendor say in favor of the cloud API, and why does the hospital’s legal team override that argument anyway?

Week 11: Anonymization & Secondary Data Use

Session objective: Students can correctly anonymize a dataset for research use.

Opening (5 min) — say this: > “You now know why data has to stay in the EU. Today’s question is different: even inside the EU, when is data ‘clean enough’ to reuse for research?”

Teaching script — anonymization vs. pseudonymization + DiGA (30 min):

- Draw a clear line between anonymization (irreversible) and pseudonymization (reversible with a key) — this distinction is where students most often make mistakes.
- Introduce the DiGA (Digital Health Applications) reimbursement pathway briefly as a real-world example of a compliance model built around exactly this kind of data governance.

Facilitating the VCS activity — anonymized extraction (3 hrs): 1. Provide a batch of Synthea records; have students extract a dataset suitable for a mock research request. 2. Require them to document every anonymization step taken — this documentation habit is the actual skill being graded, not just the end result. 3. Peer-review: swap extracts and have another student try to re-identify a record from the “anonymized” version — if they can, the extraction failed, and that’s a valuable, safe way to learn why.

Discussion facilitation (Q3.3): After the peer re-identification exercise, ask: “What did the failed extraction still contain that gave it away?” — concrete failure analysis teaches this skill better than an abstract checklist.

Week 12: Month 3 Synthesis & Checkoff

Session objective: Students produce one GDPR-compliant, anonymized care-plan data extract independently.

Opening (5 min) — say this: > “This is the last piece of Month 3 — put together everything from the last three weeks: verify the AI’s care plan, respect data sovereignty, and anonymize correctly.”

Facilitating the checkoff (graded, 3 hrs): 1. Independent task: produce one anonymized extract from a synthetic care-plan record. 2. Grade against the documentation-of-steps standard from Week 11, not just the final output. 3. Consider scheduling a break after this week if your institution observes a winter recess before Week 13 resumes.

■ MONTH 4 — ETHICOLEGAL AUTONOMY & THE FINAL CHECKOFF

Week 13: Liability Mapping

Session objective: Students can identify who bears responsibility when an AI

recommendation is wrong.

Opening (5 min) — say this: > “We’ve spent three months building technical safety skills. Today we ask the question that determines who’s actually on the hook when those skills aren’t enough: who’s liable?”

Teaching script — liability case conference (35 min):

- Present a de-identified adverse-event report (real, published, or composited from public case reports — never a real current patient case).
- Walk the class through the liability chain: the treating clinician retains the duty of independent judgment regardless of which AI tool was used — this is the legal mirror of the HITL principle from Week 1.

Facilitating the VCS activity — small-group liability mapping (3 hrs): 1.

Break into small groups; each maps the case’s liability chain (clinician / vendor / institution) and justifies the split. 2. Have groups present and compare — expect disagreement on the exact proportions, which is realistic; medico-legal liability apportionment is often genuinely contested.

Discussion facilitation (Q4.3): Push on any student who says “the AI vendor should be liable” — ask them to reconcile that with the fact that the clinician had override authority and chose not to use it.

Week 14: EU AI Act Risk Tiers

Session objective: Students can classify a clinical AI tool by EU AI Act risk tier.

Opening (5 min) — say this: > “Today is regulatory, but don’t tune out — this determines what paperwork and oversight legally attach to every tool you’ve used this term.”

Teaching script — risk tiers + current timeline (35 min):

- Present the tier structure (unacceptable / high / limited / minimal risk) and explain that diagnostic/treatment-influencing tools are the paradigm high-risk category.
- Give students the current compliance timeline honestly, including its uncertainty: the EU’s 2026 “Digital Omnibus” package pushed deadlines back — standalone high-risk systems now must comply by December 2027, and high-risk AI embedded in regulated medical devices by August 2028. Note that the underlying obligations didn’t change, only the timing — and tell students to verify against the Official Journal before treating any specific date as final, since this is a live regulatory process.

Facilitating the VCS activity — tier-classification exercise (3 hrs): 1. Have students classify every VCS tool used this term (scribe, CDSS demo, care-planning tool) by risk tier, with written justification. 2. Cross-check in pairs; disagreements should be resolved by pointing to the specific Annex criteria, not by vote.

Discussion facilitation (Q4.4): Use the “Hospital Wi-Fi” distractor option deliberately — it’s there to test whether students understand that risk tiering

applies to the AI *system's function*, not to IT infrastructure generally.

Week 15: Informed Consent & Transparency

Session objective: Students can disclose AI involvement to a patient appropriately.

Opening (5 min) — say this: > “Today isn’t a legal lecture — it’s a communication skill. You’re going to practice saying, out loud, something most clinicians have never had to say before: ‘an AI was involved in this.’”

Teaching script — consent scripting (20 min):

- Model one disclosure conversation yourself first, live, so students have heard it before they have to produce it.
- Emphasize: disclosure should be plain, non-alarming, and should name the clinician’s own verification role, not just the AI’s involvement.

Facilitating the VCS activity — standardized-patient roleplay (3 hrs): 1. Run video roleplay pairs; one plays a patient asking directly “was AI involved in my diagnosis?” 2. Peer-grade against a simple rubric: did they disclose plainly, did they explain their own verification role, did they invite questions. 3. Rotate pairs so everyone both delivers and receives the disclosure at least once.

Discussion facilitation (Q4.1): Have two or three students say their actual scripted response out loud to the class — hearing the range of acceptable phrasing is more useful than reading one “correct” answer.

Week 16: Capstone Preparation

Session objective: Students can integrate all four competency areas in one continuous case.

Opening (5 min) — say this: > “Everything until now has been one skill at a time. Today’s dry run is the first time you’ll do all four at once — and it’s ungraded on purpose, so you can find your own gaps safely.”

Teaching script — integration walkthrough (25 min):

- Walk through the full capstone rubric explicitly so there are no surprises later.
- Explicitly name the integration failure mode to watch for: a student who handles the diagnostic override well but forgets the consent disclosure, or vice versa — competence in one area doesn’t guarantee competence in another under combined pressure.

Facilitating the VCS dry-run (3 hrs): 1. Run one full ungraded simulated shift per student (or small group). 2. Provide real-time instructor feedback rather than saving it all for a written debrief — the goal this week is calibration, not evaluation.

Discussion facilitation (Q4.2): Ask students to narrate their own dry-run imaging

discrepancy (if it came up) in first person — this rehearses the exact explanation they'll need to give during the graded capstone.

Week 17: Month 4 Synthesis & Checkoff

Session objective: Students pass a combined disclosure + risk-tier classification checkoff.

Opening (5 min) — say this: > “This is the last gate before Consolidation. Two skills, graded together, because in practice you’ll never get to do them one at a time.”

Facilitating the checkoff (graded, 3 hrs): 1. One mock disclosure conversation (graded live or recorded). 2. One risk-tier classification task on a tool not previously classified in class (tests transfer, not memorization). 3. Students who fail either half repeat only that half before advancing.

■ MONTH 5 — CONSOLIDATION & MOCK OSCE

Week 18: Integration Station Design

Session objective: Students design a valid integration test case.

Opening (5 min) — say this: > “Today you become the instructor. Designing a good test case requires understanding the failure mode better than just passing a test does.”

Facilitating the activity (3 hrs): 1. Small groups design one integration station combining at least two of: scribing error, diagnostic override, sovereignty documentation, consent disclosure. 2. Groups pilot their station on a different peer group and collect feedback. 3. Circulate and coach groups whose stations are too easy or too narrowly scoped — a good station should have more than one plausible “trap.”

Week 19: Mock OSCE Round 1

Session objective: Students perform under timed, integrated pressure.

Facilitating the OSCE (3 hrs): 1. Run timed stations via video platform; score against a fixed rubric, not impression. 2. Keep strict time limits even when a student is clearly struggling — realistic time pressure is part of what’s being assessed.

Discussion facilitation (Q5.1): Debrief as a full-checkpoint walkthrough: verify transcription → question the risk score → invoke override → confirm documentation

compliance → document reasoning and consent. Have students self-score against this sequence before you reveal how they actually did.

Week 20: Mock OSCE Round 2 + Remediation

Session objective: Students needing remediation repeat stations; others receive individual feedback.

Facilitating (3 hrs): 1. Split the room: remediation students repeat stations, everyone else gets 1:1 debriefs on Round 1 performance. 2. Keep remediation non-punitive in tone — frame it as “not yet,” not “failed.”

Discussion facilitation (Q5.2): Have students generate their own example first, before you supply one, distinguishing automation bias (human failure to scrutinize) from explainability failure (system failure to justify).

Week 21: Consolidation Wrap-Up

Session objective: Full-class synthesis; readiness sign-off.

Facilitating (3 hrs): 1. Open forum — let students bring their own lingering uncertainties rather than a scripted agenda. 2. Sign off readiness individually; any student not ready gets a short, specific remediation plan before Month 6 begins.

■ MONTH 6 — SOLO PILOT CAPSTONE & CERTIFICATION

Weeks 22-23: Capstone Shifts 1 & 2

Session objective: Students run a full simulated shift independently, twice, with different case mixes.

Facilitating each shift (6 hrs each): 1. Full simulated shift: intake (AI scribing) → diagnosis (AI imaging validation) → treatment planning. 2. Grade against the Month 4 “Confidence Factor” checklist — consistency across both shifts matters more than a single strong performance. 3. Hold individual debriefs after Shift 2, specifically comparing performance across the two shifts to check for consistency, not just competence on a good day.

Week 24: Remediation / Retake Window

Facilitating: Open sandbox access; be available for targeted 1:1 practice rather than re-teaching full content. This week is about closing specific, named gaps from

the capstone shifts.

Week 25: Certification Ceremony & Course Close

Session objective: Final checkoff and course close.

Opening (10 min) — say this: > “Twenty-five weeks ago we started with one sentence: Human-in-the-Loop. Everything since has been a different way of practicing that same sentence — in documentation, diagnosis, data handling, and disclosure. That’s what you’re being certified on today.”

Facilitating (3 hrs): 1. Final graded checkoff (format per your institution’s exam policy). 2. Course evaluation. 3. Certificate ceremony — consider inviting each student to name the one skill from the course they expect to use first in real practice; it’s a strong closing ritual and gives you informal feedback on which weeks landed hardest.

General Facilitation Notes

- **Timing is a guide, not a mandate** — if a discussion is genuinely productive, let it run over and trim the next section rather than cutting it off mechanically.
- **Normalize near-misses** — several activities (Weeks 3, 7, 11) depend on students self-reporting errors honestly; if students sense punishment for admitting mistakes, the whole pedagogy breaks down.
- **Keep the HITL throughline explicit** — in every week’s opening, it helps to name which piece of the HITL principle is being extended this week (verification → override → sovereignty awareness → liability/consent → integration).
- **All VCS activities use synthetic or de-identified data only** — reinforce this rule verbally each month, not just in the syllabus.